Adaptive Differential Privacy for Federated IoHT Intrusion Detection: A Privacy-Utility Study under Heterogeneous Clients

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***Abstract*—Privacy and utility remain important concerns in federated intrusion detection for Internet-of-Healthcare-Things (IoHT) networks, especially when clients are heterogeneous and hold non-IID local data. Although federated learning (FL) reduces raw data exposure, client model updates can still leak sensitive information. Differential privacy (DP) is a widely used defense, but its utility impact can vary substantially across datasets and client conditions. In this work, we study adaptive privacy mechanisms for federated IoHT intrusion detection under heterogeneous clients. We simulate non-IID client populations using Dirichlet-based partitioning on the ECU-IoHT and WUSTL-EHMS 2020 datasets, and compare no-DP training, fixed-noise DP, smoothed adaptive noise scaling, and adaptive clipping. To quantify privacy leakage, we also implement a white-box membership inference attack and evaluate attack performance on held-out splits. Our results show that strict no-identifier preprocessing reduces membership inference performance close to chance, while identifier-inclusive WUSTL serves as a strong leakage-positive control. On the utility side, ECU remains near ceiling across privacy mechanisms, limiting the room for adaptive methods to show benefit, whereas WUSTL is substantially more sensitive to privacy perturbations. Overall, we do not observe consistent or substantial utility gains from adaptive privacy mechanisms across datasets; instead, the results suggest that their effectiveness is highly dataset-dependent and may require more challenging benchmarks or better-calibrated settings to produce meaningful improvements.**

***Index Terms*—keyword1, keyword2, keyword3**

I. Introduction

II. Related Work

III. Methodology

IV. Evaluation

V. Conclusion

Acknowledgment

References

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